Integration by Parts

ANSWER KEY

Section 1: Standard Integration by Parts Formula

1 Evaluate $\int xe^{2x}dx$.

a. $\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$ ✓

b. $\frac{1}{2}xe^{2x} + \frac{1}{4}e^{2x} + C$

c. $xe^{2x} - e^{2x} + C$

d. $\frac{1}{2}x^2e^{2x} + C$

Let $u = x, dv = e^{2x}dx \Rightarrow du = dx, v = \frac{1}{2}e^{2x}$. By parts:
$uv - \int vdu = \frac{1}{2}xe^{2x} - \frac{1}{2}\int e^{2x}dx = \frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$.

2 Evaluate $\int \ln x dx$ for $x > 0$.

a. $x\ln x - x + C$ ✓

b. $x\ln x + x + C$

c. $\frac{1}{x} + C$

d. $\frac{(\ln x)^2}{2} + C$

Let $u = \ln x, dv = dx \Rightarrow du = \frac{1}{x}dx, v = x$. By parts: $uv - \int vdu = x\ln x - \int 1dx = x\ln x - x + C$.

3 Evaluate $\int_0^1 x\cos(\pi x)dx$.

a. $-\frac{2}{\pi^2}$ ✓

b. $\frac{2}{\pi^2}$

c. 0

d. $-\frac{1}{\pi}$

Let $u = x, dv = \cos(\pi x)dx \Rightarrow du = dx, v = \frac{1}{\pi}\sin(\pi x)$.
$[\frac{x}{\pi}\sin(\pi x)]_0^1 - \frac{1}{\pi}\int_0^1 \sin(\pi x)dx = 0 + [\frac{1}{\pi^2}\cos(\pi x)]_0^1 = \frac{1}{\pi^2}(-1 - 1) = -\frac{2}{\pi^2}$.

Section 2: Tabular Integration by Parts

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4 Using tabular integration, evaluate $\int x^2 e^x dx$.

a. $(x^2 - 2x + 2)e^x + C$ ✓

b. $(x^2 + 2x + 2)e^x + C$

c. $(x^2 - 2x)e^x + C$

d. $x^2 e^x - 2e^x + C$

Differentiate $x^2 \to 2x \to 2 \to 0$. Integrate $e^x \to e^x \to e^x \to e^x$. Alternating signs: $+x^2 e^x - 2xe^x + 2e^x + C = (x^2 - 2x + 2)e^x + C$.

5 Evaluate $\int x^3 \sin x dx$ using tabular integration.

a. $-x^3 \cos x + 3x^2 \sin x + 6x \cos x - 6 \sin x + C$ ✓

b. $x^3 \cos x - 3x^2 \sin x + C$

c. $-x^3 \cos x - 3x^2 \sin x + C$

d. $-x^3 \sin x + 3x^2 \cos x + C$

Differentiate $x^3 \to 3x^2 \to 6x \to 6 \to 0$. Integrate $\sin x \to -\cos x \to -\sin x \to \cos x \to \sin x$. Product with alternating signs yields $-x^3 \cos x + 3x^2 \sin x + 6x \cos x - 6 \sin x + C$.

6 When is tabular integration applicable?

a. **When one factor is a polynomial that terminates to 0 upon repeated differentiation** ✓

b. Only for trigonometric functions

c. Only for logarithmic integrals

d. When both functions are exponential

Tabular integration is a streamlined form of repeated integration by parts, ideal when one factor is a polynomial that differentiates to zero after finitely many steps.

Section 3: Cycling Integrals (Solving for the Unknown Integral)

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7 Evaluate $\int e^x \cos x \, dx$.

a. $\frac{1}{2}e^x(\cos x + \sin x) + C$ ✓

b. $\frac{1}{2}e^x(\cos x - \sin x) + C$

c. $e^x \sin x + C$

d. $e^x(\cos x + \sin x) + C$

Integrating by parts twice reproduces $-I$. Adding I to both sides gives $2I = e^x \sin x + e^x \cos x \Rightarrow I = \frac{1}{2}e^x(\cos x + \sin x) + C$.

8 Evaluate $\int e^{2x} \sin x \, dx$.

a. $\frac{e^{2x}}{5}(2\sin x - \cos x) + C$ ✓

b. $\frac{e^{2x}}{5}(\sin x - 2\cos x) + C$

c. $\frac{e^{2x}}{4}(2\sin x - \cos x) + C$

d. $e^{2x} \cos x + C$

Integrating by parts twice produces $-\frac{1}{4}I$. Multiplying by $\frac{4}{5}$ yields $\frac{e^{2x}}{5}(2\sin x - \cos x) + C$.

9 What is $\int \sec^3 x \, dx$?

a. $\frac{1}{2}(\sec x \tan x + \ln|\sec x + \tan x|) + C$ ✓

b. $\frac{1}{3}\sec^3 x + C$

c. $\sec x \tan^2 x + C$

d. $\frac{1}{2}\sec^2 x + C$

By parts with $u = \sec x$, $dv = \sec^2 x \, dx$, cycling produces $2I = \sec x \tan x + \int \sec x \, dx$, giving $\frac{1}{2}(\sec x \tan x + \ln|\sec x + \tan x|) + C$.

Section 4: Integration by Parts with Inverse Trigonometric Functions

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10 Evaluate $\int \arctan x dx$.

a. $x\arctan x - \frac{1}{2}\ln(1+x^2) + C$ ✓

b. $x\arctan x + \frac{1}{2}\ln(1+x^2) + C$

c. $\frac{1}{1+x^2} + C$

d. $x\arctan x - \ln(1+x) + C$

Let $u = \arctan x, dv = dx \Rightarrow du = \frac{1}{1+x^2}dx, v = x$. Integral is $x\arctan x - \int \frac{x}{1+x^2}dx = x\arctan x - \frac{1}{2}\ln(1+x^2) + C$.

11 Evaluate $\int \arcsin x dx$.

a. $x\arcsin x + \sqrt{1-x^2} + C$ ✓

b. $x\arcsin x - \sqrt{1-x^2} + C$

c. $\frac{1}{\sqrt{1-x^2}} + C$

d. $x\arcsin x + \frac{1}{2}\sqrt{1-x^2} + C$

Let $u = \arcsin x, dv = dx \Rightarrow du = \frac{1}{\sqrt{1-x^2}}dx, v = x$. Integral is $x\arcsin x - \int \frac{x}{\sqrt{1-x^2}}dx = x\arcsin x + \sqrt{1-x^2} + C$.

12 Evaluate $\int_0^1 x\arctan x dx$.

a. $\frac{\pi}{4} - \frac{1}{2}$ ✓

b. $\frac{\pi}{8}$

c. $\frac{\pi}{4}$

d. $\frac{\pi}{8} - \frac{1}{4}$

By parts: $[\frac{x^2}{2}\arctan x]_0^1 - \frac{1}{2}\int_0^1 \frac{x^2}{1+x^2}dx = \frac{\pi}{8} - \frac{1}{2}[x - \arctan x]_0^1 = \frac{\pi}{8} - \frac{1}{2}(1 - \frac{\pi}{4}) = \frac{\pi}{4} - \frac{1}{2}$.

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